# HWRS 561b Physical Hydrogeology II
Spring 2027

## Instructor and Contact Information

Ty Ferré
Lecture time: M/W/F, 2:00 to 2:50 pm MST (Arizona time); classroom to be announced.
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## Course Description

This course is the second part of Physical Hydrogeology and focuses on the physical principles governing water flow and solute transport in variably saturated porous media. The course approaches these topics from a fundamental multiphase flow and transport perspective, with an emphasis on developing strong physical intuition. Building on this intuitive foundation, we formulate mathematical models and examine the physical behavior of solutions describing variably saturated flow and transport processes. The course also introduces selected cutting-edge topics related to the fate and transport of emerging contaminants in the subsurface, such as PFAS.

## Course Prerequisites or Co-requisites

No pre-requisites

Must be enrolled in the MS Hydrogeology program

Required co-registration in HWRS 562b Chemical Hydrogeology II, HWRS 563b Hydrogeologic Measurement Methods II, HWRS 564b Hydrogeologic Analysis Tools & Methods II, and HWRS 565b Communications in Hydrogeology II

## Required Textbooks/Materials

Selected chapters from the following textbooks will be used in this course. Scanned copies of the required chapters will be provided.

- *Contaminant Hydrogeology* — C. W. Fetter, T. Boving, and D. Kreamer (2018)
- *Subsurface Hydrology* — G. F. Pinder and M. A. Celia (2006)
- *Soil Physics* — W. A. Jury and R. Horton (2004)

## Course Objectives

Students will

1. Gain an understanding of the physical concepts and pore-scale controls underlying water flow and solute transport in variably saturated media.
2. Gain an understanding of the assumptions behind macroscopic governing equations for flow and transport in variably saturated media.
3. Gain an understanding of the analytical and numerical methods for solving flow and transport equations in variably saturated media.

4. Be able to apply analytical and numerical solutions for flow and transport in variably saturated media to analyze problems under a wide range of conditions to gain physical intuition.
5. Be able to apply concepts and tools to analyze contaminant transport in the vadose zone.

## Expected Learning Outcomes

Students will be able to

1. Characterize saturation-dependent soil hydraulic properties, both qualitatively and quantitatively, and develop an understanding of the relationship of those properties with pore scale air and water distributions.
2. Develop appropriate physical and mathematical representations of water flow and solute transport through fully and partially water saturated porous media at multiple scales in 1D, 2D, and 3D.
3. Synthesize and contextualize knowledge of data collection methods, quantitative models of flow and solute transport, and fundamental conceptual understanding for the purposes of supporting water resources decision-making.

## Course Format and Teaching Methods

The course will consist of a mixture of lectures, in-class discussion and exercises.

## Planned Field Trips

The first week of Spring 2027 (January 13 to 15) is reserved for the MS Hydrogeology field trip, which applies to this course and the co-convened courses HWRS 562b, HWRS 563b, HWRS 564b, and HWRS 565b. There will be no classroom meetings on January 13 or 15.

## Schedule of Topics & Activities

The course will be organized around month-long projects that provide a unifying learning context across all five co-convened classes. It follows a theory–data–prediction framework, combining case-based theoretical development with project-based methodological practice. The scheduled activities are outlined below. The Spring 2027 schedule below uses Arizona time (MST). Topic pacing, homework dates, and the oral midterm date are tentative. Regular classroom meetings begin January 20 and end May 5. There are no classes on January 18 or during spring break, March 6 to 14. Reading Day is May 6.

| Week | Date (2027) | Topic or activity | Homework (tentative; MST) |
|---|---|---|---|
| 1 | Wed Jan 13 | Field trip; no classroom meeting (January 13 and 15). | |
| 2 | Mon Jan 18 | Martin Luther King Jr. Day; no classes. | |
| **Module 1: Solute transport in saturated media** | | | |
| 2 | Wed Jan 20 | Conservative solute transport: Pore-scale controls, macroscopic formulation, governing equations, BCs, 1D Taylor-Aris dispersion | |
| 2 | Fri Jan 22 | Conservative solute transport: Pore-scale controls, macroscopic formulation, governing equations, BCs, 1D Taylor-Aris dispersion | |
| 3 | Mon Jan 25 | Conservative solute transport: Analytical solutions, breakthrough curves. | HW 1 assigned.<br>Due Feb 4, 10 pm. |
| 3 | Wed Jan 27 | Conservative solute transport: Analytical solutions, breakthrough curves. | |
| 3 | Fri Jan 29 | Conservative solute transport: Analytical solutions, breakthrough curves. | |
| 4 | Mon Feb 1 | Non-conservative solute transport: retardation, kinetic/equilibrium adsorption | |
| 4 | Wed Feb 3 | Non-conservative solute transport: Analytical solutions, breakthrough curves. | |
| 4 | Fri Feb 5 | Non-conservative solute transport: Analytical solutions, breakthrough curves. | |
| 5 | Mon Feb 8 | Non-conservative solute transport: Analytical solutions, breakthrough curves. | HW 2 assigned.<br>Due Feb 18, 10 pm. |
| **Module 2: Pore-scale controls and macroscopic description of variably saturated flow** | | | |
| 5 | Wed Feb 10 | Pore-scale fluids distribution (capillary tubes). | |
| 5 | Fri Feb 12 | Pore-scale fluids distribution (capillary tubes) | |
| 6 | Mon Feb 15 | Pore-scale fluids distribution (capillary tubes) | |
| 6 | Wed Feb 17 | Macroscopic descriptions of air/water distribution | |
| 6 | Fri Feb 19 | Macroscopic descriptions of air/water distribution | |
| 7 | Mon Feb 22 | Macroscopic descriptions of air/water distribution. | HW 3 assigned.<br>Due Mar 4, 10 pm. |
| 7 | Wed Feb 24 | Macroscopic descriptions of air/water distribution. | |
| **Module 3: Steady-state and transient variably saturated flow** | | | |
| 7 | Fri Feb 26 | Two-phase flow and Richards Equation. | |
| 8 | Mon Mar 1 | Two-phase flow and Richards Equation. | |

| Week | Date (2027) | Topic or activity | Homework (tentative; MST) |
|---|---|---|---|
| 8 | Wed Mar 3 | Steady-state 1D unsaturated flow. | |
| 8 | Fri Mar 5 | Steady-state 1D unsaturated flow. | |
| 9 | Mon Mar 8 | Spring break; no classes March 8, 10, or 12. | |
| 10 | Mon Mar 15 | Transient 1D unsaturated flow. Scenario analysis. | |
| 10 | Wed Mar 17 | Midterm review and discussion | |
| 10 | Fri Mar 19 | Midterm exam (oral, tentative) | |
| 11 | Mon Mar 22 | Transient 1D unsaturated flow. Simplified models (Green-Ampt, Philip's models). | HW 4 assigned.<br>Due Apr 1, 10 pm. |
| 11 | Wed Mar 24 | Transient 1D unsaturated flow. Simplified models (Green-Ampt, Philip's models). | |
| 11 | Fri Mar 26 | Transient 1D unsaturated flow. Numerical solutions of Richards equation. | |
| 12 | Mon Mar 29 | HYDRUS-1D tutorial and preparation exercise | |
| 12 | Wed Mar 31 | In-class HYDRUS-1D exercise. | |
| 12 | Fri Apr 2 | Inverse modeling using HYDRUS-1D | |
| 13 | Mon Apr 5 | Inverse modeling using HYDRUS-1D | HW 5 assigned.<br>Due Apr 15, 10 pm. |
| 13 | Wed Apr 7 | Unsaturated flow in multidimensions. | |
| 13 | Fri Apr 9 | Solute transport with unsaturated flow. | |
| **Module 4: Risk assessment of contaminant transport in the vadose zone and groundwater** | | | |
| 14 | Mon Apr 12 | Transport of PFAS in the vadose zone. | |
| 14 | Wed Apr 14 | Transport of PFAS in the vadose zone. | |
| 14 | Fri Apr 16 | Transport of PFAS in the vadose zone. | |
| 15 | Mon Apr 19 | Transport of PFAS in the vadose zone. | HW 6 assigned.<br>Due Apr 29, 10 pm. |
| 15 | Wed Apr 21 | Transport of PFAS in the vadose zone. | |
| 15 | Fri Apr 23 | Transport of PFAS in the vadose zone. | |
| 16 | Mon Apr 26 | Transport of PFAS in the vadose zone. | |
| 16 | Wed Apr 28 | Transport of PFAS in the vadose zone. | |
| 16 | Fri Apr 30 | Transport of PFAS in the vadose zone. | |
| 17 | Mon May 3 | Course review | |
| 17 | Wed May 5 | Course review | |
| 17 | Thu May 6 | Reading Day; no classes. | |
| 17 | Fri May 7 | Final exam, 1:00 to 3:00 pm MST. | |

## Course Assessments and Grading Breakdown

You will be assessed on the basis of weekly assignments. You will also be assessed based on how you apply the understanding gained in this class to the projects. Finally, you will receive completion credit for completing weekly self-assessments.

**Assignments** are assigned Monday, and they are due on Thursdays by 10 pm. Each assignment will be given a time budget. This will give students an indication of the level of effort expected for each assignment. Students will report the time spent on the project. If individual students are spending more time than the average, the instructor will meet with them to identify and solve any issues. If many students are spending more than the assigned time, then the assignments will be modified for the remainder of the term and for subsequent years. It is expected that students will spend no more than 2.5 hours per week on these class assignments.

**Projects** are assigned monthly. These projects are designed to synthesize the content/skills you are learning in all 5 courses that month to address a hydrogeologic problem/task. This means that the projects are not specific to any one course. However, you will be assessed on the content/skills related to each specific course separately within the single project. To do this, a rubric filled with criteria as it applies to each hydrogeologic topic/course will be used. You will receive a score of either mastery, proficient, basic, or below basic for each project in each of the 5 areas:

- physical hydrogeology
- chemical hydrogeology
- hydrogeologic measurement methods
- hydrogeologic analysis tools and methods
- communication in hydrogeology

The grade you receive for the physical hydrogeology skills/content portion of the projects will be used to calculate your total course grade in Physical Hydrogeology II for the semester.

**Class participation** will be earned by participating in designated class discussions.

The percentage distribution of your grade will be as follows.

| | |
|---|---|
| Course Assignments | : 45% |
| Projects – physical hydrogeology component | : 10% |
| Mid-term exam | : 20% |
| Final exam | : 20% |
| Participation | : 5% |
| Art of porous media flow | : 5% (bonus) |

University policy regarding grades and grading systems is available at this link.

## Examinations

There are midterm and final examinations in this course. The midterm exam will be oral, and the final exam will be written. Both will be in-person. The oral midterm is tentatively scheduled for March 19. The final exam is scheduled for Friday, May 7, 2027, from 1:00 to 3:00 pm MST. A review session will be scheduled before both exams. In addition, students will be completing projects this semester that cut across all courses in the MS Hydrogeology program, and will require students to utilize and synthesize the skills they learned in all 5 courses to address a hydrogeologic question/problem.

## Grading Scale

Your final grade will be informed via D2L. Letter grades are determined using the following scale:

A: >= 90%

B: >= 80 - 89%

C: >= 70 – 79 %

D: >= 60 to 69 %

E: below 59 %

University policy regarding grades and grading systems is available at https://catalog.arizona.edu/policy/courses-credit/grading/grading-system.

## Late work Policy

No late work will be accepted for a grade, but students will receive feedback on all submitted work.

## Incomplete (I) or Withdrawal (W):

Requests for incomplete (I) or withdrawal (W) must be made in accordance with University policies, which are available at this link associated with the registrar.

## University of Arizona Course Policies

All University of Arizona course and syllabi policies, as well as other helpful information and resources, can be found at this link.

If you are in need of basic needs care, here is another helpful link, in addition to what you can find at the policy link above.

## Subject to Change Statement

Information contained in the course syllabus, other than the grade and absence policy, may be subject to change with advance notice, as deemed appropriate by the instructor.